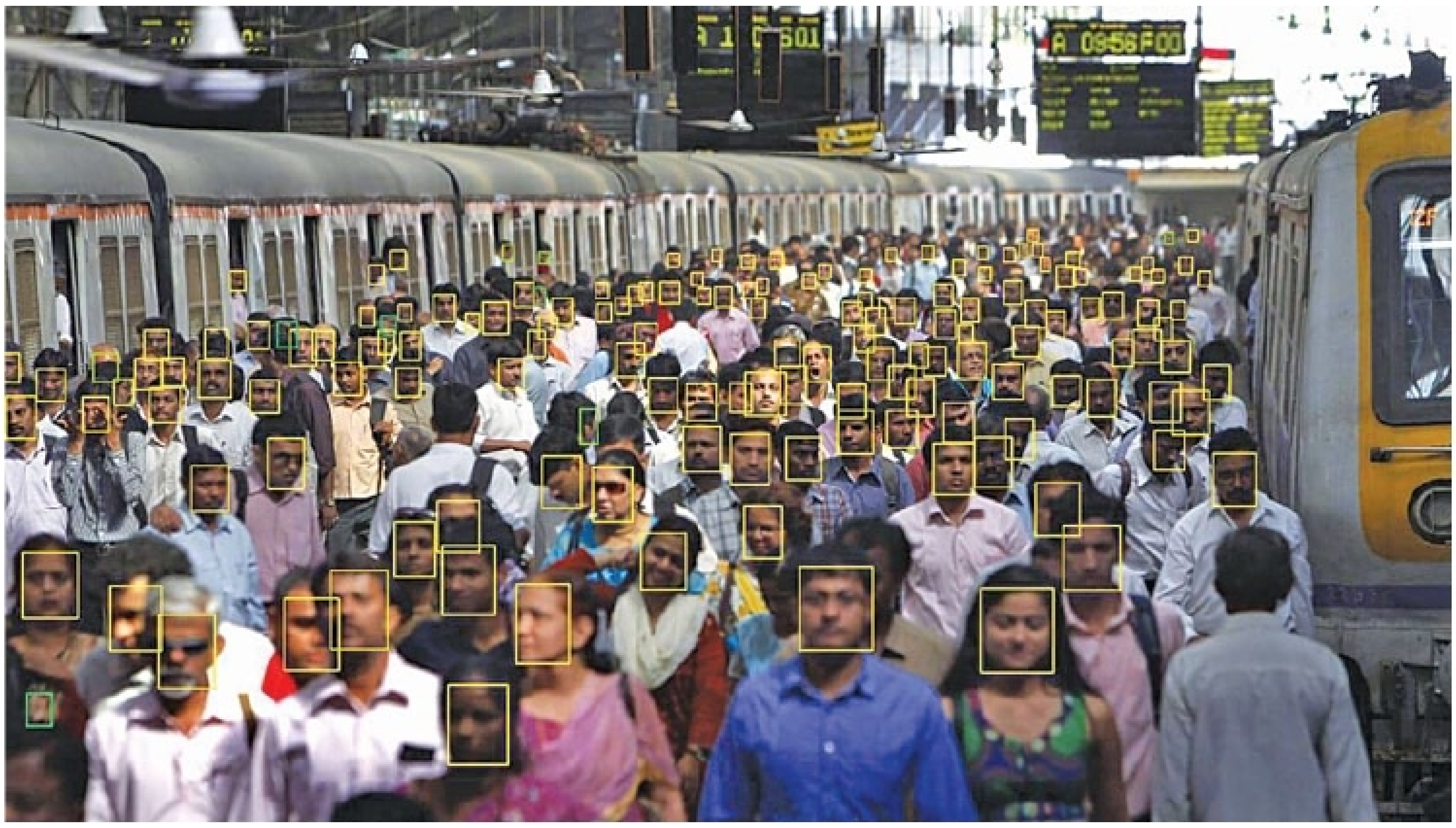
STARTUP ZONE

1 LARGE-SCALE FACE DETECTION By AI-Powered Streetlights



Inkers in collaboration with Hynetic's smart streetlights has successfully deployed an artificial intelligence (AI)-powered solution that works in real time. Intersection of the Internet of Things (IoT) and AI combined with distributed computing frameworks allow for a feature-rich implementation of a technology that is highly optimised for safety while ensuring significant energy savings. It aims to simplify AI adoption and remove current bottlenecks like scalability, data security and privacy, support for multiple AI application on the same hardware and cost of deployment.

The system on real-time basis classifies moving objects like humans, vehicles, animals or no objects, and adjusts the intensity of the light either by increasing or decreasing it to optimise energy usage. In a nutshell, lighting automatically dials down the brightness when no one is around and restores radiance upon detecting the presence of oncoming activity. With smart lighting, overall savings run into 50 to 60 per cent of total electricity consumption, which runs into millions of rupees.

Application of the technology includes motion estimation, feature extraction, action and scene recognition, faces go-to emotion-detection model and newly-seen features to zero-shot detector. All stages help create heavily compressed videos and a fully-segmented global model. The system can count people and detect faces with above 95 per cent accuracy for large crowds and rallies. It detects and tracks the same objects on multiple cameras in real time or offline videos. It is specially trained for Indian faces, and can achieve up to five times facial super-resolution. It can search through millions of embeddings with the in-GPU database in real time.

It has dense 3D reconstruction with VR deployment for buildings, landscapes and other facilities. Extremely-accurate sparse 3D reconstruction helps in fast and reliable 3D map creation. The system can learn to detect and track new objects without prior training, for immediate deployment.

Speaking on the collaboration, Manish Giri, co-founder, Inkers Tech-